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Nobody

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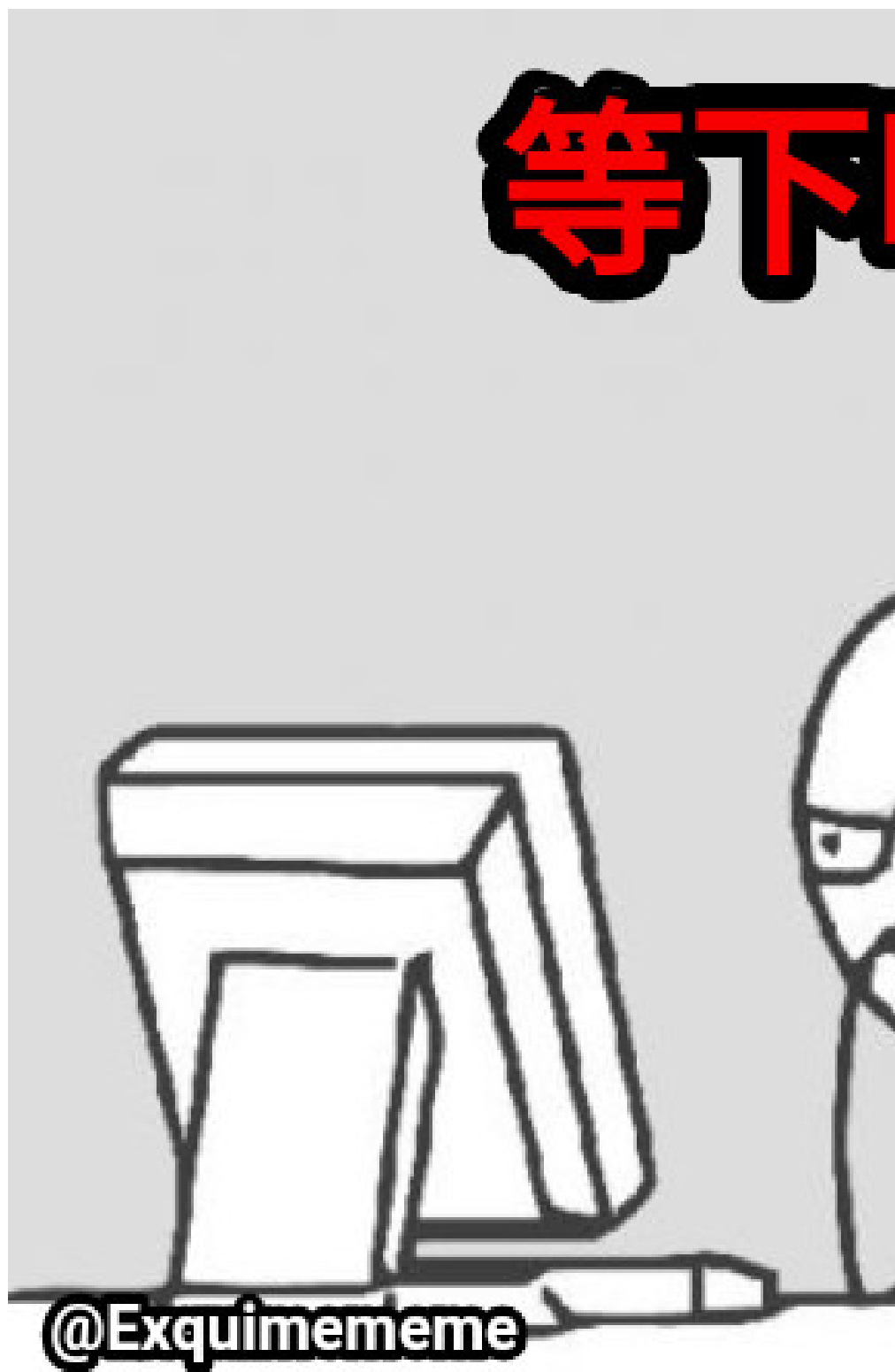
1 index

Disclaimer : this is a story with full of malaysia slang...

2 Chapter 1: The Biggest Git Conflict in Math History

Long long time ago... back in the 1600s....

The biggest difference between people back then and people nowadays comes down to what occupies our minds,



People today: "*Later eat what?*"

People in the 17th century: *“How exactly did God design the universe?”*

Back then, religion was a massive deal in Europe. Most scientists truly believed that the universe was not a random accident. Instead, they thought God had carefully architected it.



Those scientists seemed to share one ultimate endgame quest:

To figure out how the universe actually works.

Why don't planets just randomly propel themselves into deep space?

Why is the Moon just “sims” after the Earth

And why do planets orbit the Sun with what looks like suspiciously perfect stability?



If you could answer these questions, Yay ! —you had cracked the source code of the universe.

At that time, many scientists were also had a heavy religious background, so studying nature wasn't just science. It was basically trying to peek into God's dev tools without triggering a permanent ban.



And whoever managed to crack that system first wasn't just a scientist anymore. They were essentially the ultimate reality hackers—the system administrators of physics itself.

However, if you want to crack a source code, you need a test file to run experiments on.

Luckily for them, they didn't have to look far. They just looked up at the sky and found a massive dataset already running flawlessly in production: planetary motion.

3 Chapter 2:

That's when one of the biggest architecture debates in early science broke out.

Was the universe running on an Earth-centered model? Or a Sun-centered model?



In other words, was Earth the center of the entire system, or just another local node orbiting a much larger compute unit?

Back then, those who are supporting the Sun-centered model were basically getting roasted every single day. Case in point, an Italian philosopher Giordano Bruno, also a mathematician, was “roasted” alive at the stake in Rome’s market square, just because he took the Sun-centered idea and used it to aggressively troll the religious establishment.



The Earth-centered model had been patched and maintained by generations of legacy scholars for over a thousand years. Even though it sucks with extremely bloated code, it still ran..... more or less.

On the other hand, the Sun-centered model looked like a much cleaner, more elegant architecture. But there was a catch: you had to prove it with reliable evidence. You had to take your new system and accurately predict exactly where a planet would be next.

Under this brutal test dataset, the scientific world did not believe in tears, and they didn't give a fxxk about a beautiful Canva presentation. What mattered was the real output.

If you could accurately predict where a planet would be in the next moment, you were the strongest person on Earth. If you couldn't, then no matter how beautifully you presented the Sun-centered model, it was still just unoptimized hype.

Just when everyone was getting completely burned out by these brutal benchmark tests, a German astronomer named Kepler stepped in.

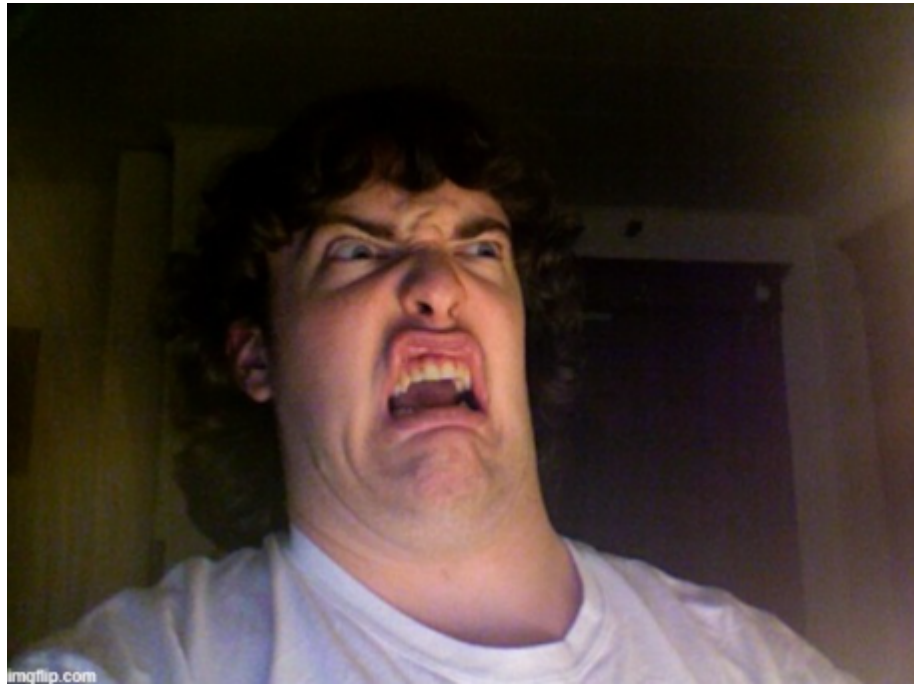
4 Chapter 3:xxxx

He was the ultimate grind machine of science—the kind of developer who doesn’t sleep, doesn’t rest, and can spend six straight years obsessing over something as chaotic as Mars’ orbit. By filling thousands of pages of scratch paper, he managed to push the new architecture forward in a critical way.

He eventually formulated the famous laws of planetary motion. His second law basically stated this: planets sweep out equal areas in equal amounts of time.

Suddenly, those supporting the Sun-centered model felt like the sun had finally come out. The core logic worked; the architecture held.

But just as mathematicians excitedly spread out their papers, ready to turn this into a clean predictive formula, the room collectively hit a “smile to soul-exit” moment.



Wait, what? The orbit... is an ellipse? imgflip.com

This irregular, curved shape... where exactly is the formula supposed to come from?

Because mathematical tools at the time were severely underdeveloped, a whole pile of long-standing legacy bugs started flooding everyone’s To-Do list:

- 1.Find why planets move in ellipses
- 2.Invent new math (Develop Calculus)
- 3.Crack “Kepler Equation”
- 4.The Multi-Body Problem

.....

Back then, the table of contents of a thick geometry textbook was basically what drove countless brilliant minds to exhaustion—a “10,000 Whys” problem set in disguise.

It wasn’t that people weren’t smart enough. It was simply that they didn’t have the right programming tools to work with.

Geometry and algebra at the time were mostly static systems, while the world they were trying to describe was full of dynamic, real-time change. Trying to handle that with existing tools was like using a plastic toy shovel to dig the Mariana Trench. It just wasn’t built for the job.

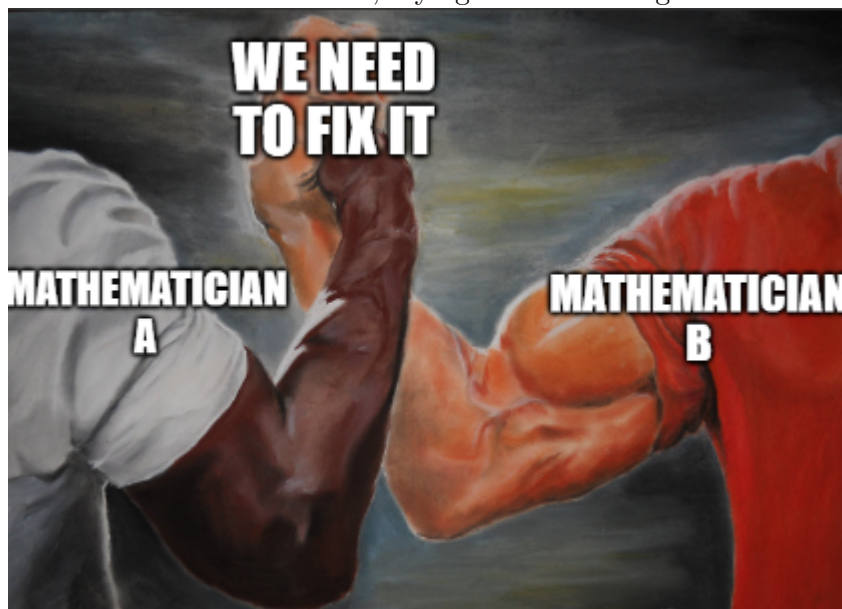
So, the entire European scientific community got stuck in place, collectively screaming for a god-tier productivity tool that could finally handle change over time.

5 Chapter 4

In an era when even calculators didn't exist, the idea that a single lone genius could suddenly "have a flash of insight" and save the world was simply unrealistic.

Faced with this universe-level bug that had effectively stalled all of humanity, the entire European mathematical community quietly formed a massive, decades-long "grind squad" and started brute-forcing the problem.

Since solo attempts weren't working, the top "hackers" of Europe naturally shifted into an open-source collaboration mode. Everyone took on different tasks, frantically contributing modules to the shared codebase, trying to build this god-tier tool through pure crowd



power.

Fermat focused on one direction and essentially extracted the "tangent module" by force.

Descartes rebuilt the foundation layer and shipped analytic geometry as a core system upgrade.

Kepler and Cavalieri acted as frontline stress testers, pushing hard against the concept of infinitesimals.

Barrow and Wallis kept pushing continuous contributions and iterative patches.

It was like everyone was jointly assembling an enormous, impossible jigsaw puz-



zle.

By the mid-17th century, everyone rubbed their red, sleep-deprived eyes and suddenly realized in shock.

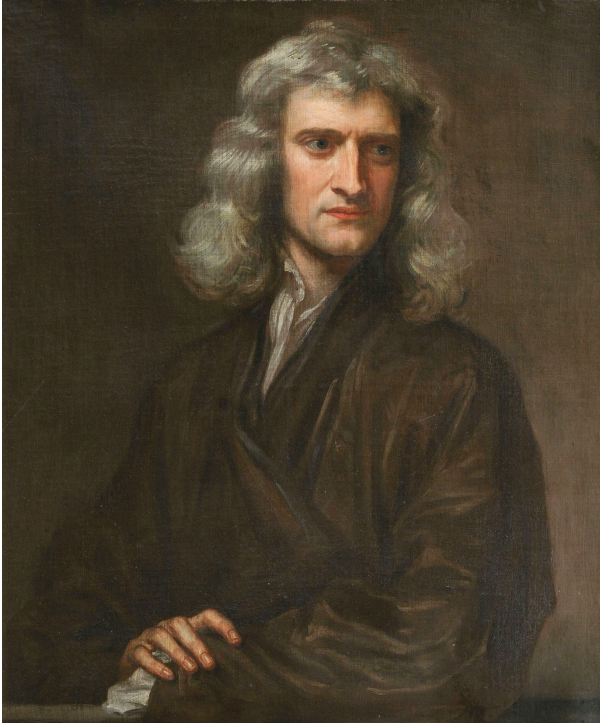
Wow The puzzle was already about 99% complete.

Everything was in place. The isolated algorithm modules were already built. All that was missing was the final step, integrating everything into a unified calculus framework.

And the next two characters to enter the scene would each, in their own genius and rivalry-driven way, finally commit the last one percent of the code.

They were

Isaac Newton,



the backend overlord of England's academic world,
and Gottfried Wilhelm Leibniz,



the full stack frontend genius from Germany.

6 Chapter 5

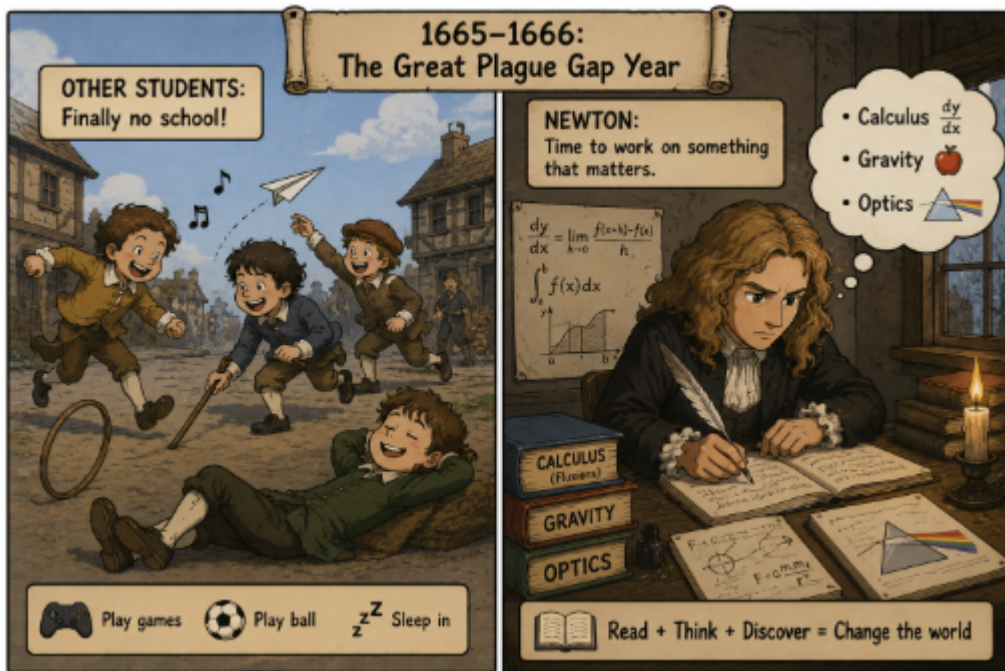
In the hot, tense summer of 1665, a devastating guest arrived on the shores of England.

The Great Plague was sweeping through London, turning bustling streets into ghost towns marked with red crosses.

Seeking safety, Cambridge University packed up its faculty and sent its students packing.

Among those fleeing the contagion was a quiet, somewhat eccentric 22-year-old student. While most of his peers treated the forced hiatus as an unexpected vacation—or a miserable pause on their futures—this young man packed his notebooks, headed back to his family’s isolated manor at Woolsthorpe, and casually embarked on what would become the most ridiculously productive gap year in human history.

His name was Isaac Newton.



Newton viewed quantities as continuously flowing. He called a changing quantity a Fluent, and its rate of change a Fluxion.

As you can see, he had already established most of the foundations of calculus. But Newton had one fatal flaw: He hated publishing. After finishing his work, he simply locked the manuscripts in a drawer and moved on to other research.

7 Chapter 6

While Newton’s manuscripts were still sitting in his drawer, a German mathematician named Gottfried Wilhelm Leibniz appeared.

At the time, he had no knowledge that Newton had already developed a similar method, so he independently worked it out himself.

Most importantly, Leibniz not only developed calculus independently, but also published his work immediately. He also introduced an elegant notation:

$$\frac{dy}{dx} \quad \int$$

These symbols are still used throughout the world today. You could say that history experienced a giant Git conflict: two great mathematicians from different countries submitted their own calculus projects almost simultaneously.

Newton wrote the core engine, while Leibniz independently built another implementation and designed the “UI” that mathematics classrooms around the world still use today.

8 Chapter 7

Decades passed before Newton suddenly blinked, looked up from his experiments, and realized:

“Wait... why does everyone in Europe think Leibniz invented calculus?”

The British mathematical community instantly rallied behind their hometown hero, fiercely whispering, “*Leibniz must have snuck a peek at Newton’s secret manuscripts! He’s a plagiarist!*”

The German side fired back with equal venom: “*He discovered it completely independently! Your guy kept his math in a drawer; you don’t get to claim the crown now!*”

The mudslinging escalated until it threatened to tear European science apart. To stop the drama, the Royal Society of London stepped in. They promised to run a fair, official investigation to see who actually invented calculus first.

They assembled an elite committee to meticulously “examine the sincerity of the Originals and compare the Copies therewith.” They promised a fair trial based purely on the historical timeline.

But there was one hilarious and unbelievable problem.

The leader of the Royal Society—the man who picked the committee members and secretly wrote the final report—was none other than Issac Newton himself.

Yes, you read that right! Newton acted as the judge, the jury, and the witness all at the same time. It is the most famous case in history of “investigating yourself.”

Of course, after a “careful” review, Newton’s committee decided that Isaac Newton was the true winner and the only inventor of calculus. The rest of Europe could only stare in shock at this totally unfair trial.

So, who really won?

Today, history experts agree that both men were geniuses who figured out calculus completely on their own. Newton did it first in secret, but Leibniz was the one who actually shared it with the world.

9 Chapter 8

Even though calculus was finally out in the open, mathematicians soon realized something was wrong. There was a major bug in the system

.When finding a derivative, you had to use tiny, near-zero numbers called infinitesimals :

$$\Delta x$$

At the start of the math problem, these tiny numbers could not be zero, because you cannot divide by zero. But mysteriously, by the end of the problem, those same numbers just seemed to disappear into thin air.

An Irish philosopher and bishop named George Berkeley stepped forward and looked at the math. He asked a few simple, devastating questions:

“Is Δx actually zero? If it isn’t zero, why does it disappear? If it is zero, how can you divide by it?”

Berkeley famously mocked these vanishing numbers, calling them the “**Ghosts of departed quantities.**”

The mathematical community fell completely silent. No one had a good answer. This officially started the **Second Crisis in the Foundations of Mathematics**. Everyone realized that while calculus gave the correct answers, it lacked a solid, logical foundation.

Eventually, the next generation of mathematicians would step up to fix this bug. They would finally make calculus safe and logical using a new concept **limits** .

10 Chapter 9

For a long time, nobody knew how to fix Bishop Berkeley's ghost bug.

The foundations of calculus were completely broken, and mathematicians were just crossing their fingers and hoping their equations wouldn't collapse. Solo attempts to solve the mystery weren't working.

Eventually, the top math "hackers" of Europe naturally shifted into an open-source collaboration mode.

Everyone took on different tasks, frantically contributing modules to the shared codebase, trying to build a god-tier logical tool through pure crowd power. They realized they finally needed to work together.

Brilliant minds like Augustin-Louis Cauchy and Karl Weierstrass stepped up to the keyboard. They decided to throw away the confusing idea of "ghost numbers" entirely.

Instead of asking what happens when Δx *becomes* zero, they asked: *What happens as Δx gets closer and closer to zero, without ever actually touching it?*

With this single, genius pivot, they created the formal definition of **limits**.

By working together and combining their braincells, the open-source mathematical community successfully patched the bug.

They gave calculus the rock-solid, logical foundation it always deserved, turning Newton and Leibniz's wild ideas into the most powerful tool in modern science.

11 outro

Today, the calculus we learn looks like just another textbook.

But it is actually the culmination of more than three hundred years of relentless effort by countless mathematicians, physicists, and astronomers.

It wasn't the result of Newton having a sudden stroke of genius one day, nor was it the work of Leibniz alone.

It was a magnificent relay race run by generations of scientists.

Newton and Leibniz simply stood on the shoulders of all the giants before them to take that final, monumental step.

Because of calculus, humanity possessed for the very first time a precise language to describe motion, change, and the laws of nature.

From the orbits of planets to the launch of rockets; from electromagnetic fields to the optimization algorithms behind modern Artificial Intelligence—almost all of modern science is inseparable from calculus.

As the saying goes:

“If mathematics is a language, then calculus is humanity’s very first grammar book for reading the universe.”